

MUHAMMAD JUNAID

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EDUCATION

Government College University Faisalabad (GCUF) <i>Bachelor of Science in Computer Science (BS CS); CGPA: 3.80 / 4.00</i> • Affiliated Campus: Govt. Graduate College, Adhiwal chowk, Jhang • Relevant Coursework: Artificial Intelligence, Computer Networks, Database Systems, Distributed Computing, Data Structures & Algorithms, Operating Systems, Software Engineering.	Expected Graduation: August 2027 <i>Punjab, Pakistan</i>
Govt. Graduate College, Adhiwal Chowk, Jhang <i>Higher Secondary School Certificate (HSSC – F.Sc Pre-Engineering; Marks: 850 / 1100)</i>	2021 – 2023 <i>Punjab, Pakistan</i>
Govt. Boys Secondary School Rail Bazar, Jhang Sadar, Jhang <i>Secondary School Certificate (SSC) – Science; Marks: 1071 / 1100</i>	2019 – 2021 <i>Punjab, Pakistan</i>

RESEARCH & ENGINEERING PROJECTS

Interactive 3D Virtual Simulation System (Undergraduate Capstone) • Built an interactive 3D web simulation that allows students to conduct virtual chemistry experiments with real-time visual feedback and multi-angle views. • Integrated reaction calculation logic to accurately determine titration outcomes, displaying live data charts and saving experiment results to a backend dashboard.	Three.js, React, MongoDB, Node.js
Autonomous Quadruped Robot & Embedded AI System • Built a physical 8-DOF quadruped robot using an ESP32-S2 board, wiring power circuits and adapting open-source gait control algorithms in C++ for stable walking. • Created an asynchronous Python companion application connecting the robot to the Groq LLM API, enabling real-time voice commands, speech synthesis, robot motion and interactive conversational persona.	ESP32-S2 Mini, C++, Python, Groq API
NotesNest – Full-Stack Note Management Platform • Developed a full-stack note-taking platform featuring secure user authentication (JWT), category filtering, reminder and rich-text editing. • Built REST API endpoints in Node.js/Express connected to a MongoDB Atlas cloud database to store, update, and persist user notes in real time.	MongoDB, Express.js, React, Node.js, Tailwind CSS

TECHNICAL SKILLS & FOCUS AREAS

Applied AI & Integration: LLM API Integrations (Groq, Gemini), Prompt Engineering, Voice / TTS Pipelines, Python

IoT & Embedded Systems: ESP32-S2, C/C++, Firmware, Servo Calibration, Circuit Prototyping

Web & Backend Development: Node.js, Express.js, REST APIs, MongoDB Atlas, React, Tailwind CSS

Tools & Platforms: Git, GitHub, Linux, Visual Studio Code, Postman, Vercel / Render Deployment

ACADEMIC HONORS & CERTIFICATIONS

- **Punjab Educational Endowment Fund (PEEF) Scholarship:** Merit scholarship awarded for top marks in FSc Pre-Engineering.
- **Google AI Professional Certificate:** Applied AI workflows, prompt design, and intelligent systems (In Progress).
- **Google AI Essentials:** Machine learning fundamentals and generative AI models (Coursera).

LANGUAGES

English: Professional Working Proficiency (Medium of Instruction Certificate available upon graduation)

Urdu: Native